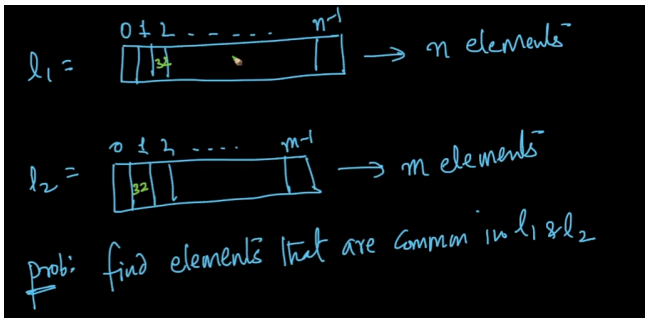
Misc

02 January 2024 00:11



We'd like to find the time complexity.

Say m < n.

Simple, brute force approach:

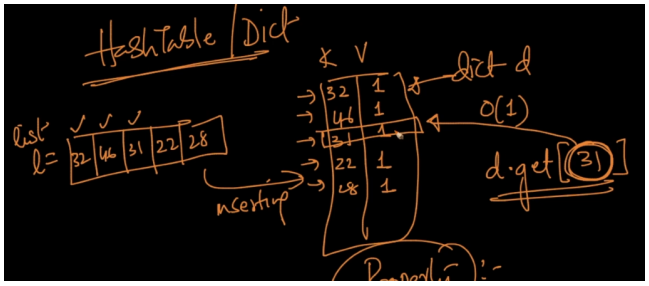
Take 0th element of the larger list (n)
It'll be the query element.
Check q in all of the other list (m elements)
For 1st element, m comparisons.
For 2nd element, m comparisons.
And so on

m+m+m+m+…+ n times = O(n*m)

T = O(n*m)
Space = O(1)

Better algo:

We could use a dictionary(hash table).



Dictionaries give O(1) access time, so on giving a key, we can search it.

So what we'll do is take elements of the smaller list (m) in a dictionary.
Space used = O(m), time = O(m) to insert elements.

Then we'll go through n elements (larger) one by one and check in the dictionary.
Time = O(n)

So total time = O(n+m)
And m <= n
T = O(n)
Space = O(m)